Individual Student Contribution Summary — Student 2

Student Name: Anshul Bhusal
Institution: MIT Academy of Engineering (MITAOE)
Project Title: NetSage AI: Automated Cisco Packet Tracer Diagnostic & Troubleshooting Framework
Assigned Project Role: AI Diagnostic Prompt Engineer & Dataset Curator

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1. Executive Summary of Contributions

My primary responsibility was designing the structured AI diagnostic system prompt, defining the strict JSON response schema, curating the master dataset (`cases.csv`), running AI diagnostics across all 20 fault cases, and building the reference case benchmark suite (`CASE-013` to `CASE-020`).

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2. Detailed Task Breakdown & Deliverables Owned

Task 1: AI Diagnostic System Prompt Engineering (`diagnose_prompt.md`)

- Developed the system prompt document `diagnose_prompt.md` .
- Established core principles for AI networking diagnostics:
  - **Strict Grounding:** Base all conclusions exclusively on provided CLI evidence.
  - **Fact vs. Inference Separation:** Distinguish clearly between observed CLI facts and logical inferences.
  - **No Hallucination:** Prevent inventing non-existent interfaces, subnets, or ping results.
- Enforced a strict JSON response schema for all AI outputs:

```
{
  "root_cause": "Concise statement of identified root cause",
  "confidence": "High | Medium | Low",
  "evidence": ["Factual quote 1", "Factual quote 2"],
  "next_command": "Decisive next command to run if confidence is lim
  "fix_steps": ["Step 1: CLI command", "Step 2: CLI command"]
}
```

- Included 3 complete worked diagnostic examples covering Layer 1 interface down, Layer 3 gateway mismatch, and Layer 2 missing VLAN.

Task 2: Master Case Dataset Curation (`cases.csv`)

- Created and populated the master CSV dataset `cases.csv` containing 21 rows (1 baseline + 20 fault cases) across 23 detailed metadata fields.
- Formatted fields including `case_id`, `symptom`, `topology_note`, `initial_configuration_or_context`, `textual_evidence`, `commands_run`, `expected_fault`, `confirmed_root_cause`, `osi_layer`, `concept`, `severity`, `ai_root_cause`, `ai_confidence`, `ai_evidence`, `ai_next_command`, `ai_fix_steps`, `human_review_decision`, `human_correction_or_notes`, `final_fix_steps`, `verification_method`, `verification_result`, `case_status`, and `evidence_status`.

Task 3: Extended AI Reference Benchmark Cases (`CASE-013` to `CASE-020`)

- Designed 8 advanced reference benchmark cases in `cases.csv` to evaluate AI performance on complex enterprise network failures:
  - `CASE-013` **(Layer 4):** Extended ACL ICMP deny rule filtering ping packets while leaving HTTP active.
  - `CASE-014` **(Layer 3):** NAT overload access-list configured with wrong internal subnet (`10.0.0.0/8` instead of `192.168.1.0/24`).
  - `CASE-015` **(Layer 7):** Unassigned default gateway IP (`192.168.10.254`) in router DHCP pool.
  - `CASE-016` **(Layer 2):** Port security violation placing switch access port into `err-disabled` state.
  - `CASE-017` **(Layer 1):** Duplex mismatch causing late collisions (`S1 Fa0/10 half-duplex` vs `R1 Fa0/0 full-duplex`).
  - `CASE-018` **(Layer 7):** Client DNS server IP set to unreachable address (`192.168.1.250`).
  - `CASE-019` **(Layer 3):** OSPF Router-ID collision (`1.1.1.1` on both routers) generating `%OSPF-4-DUP_RTRID` syslog logs.
  - `CASE-020` **(Layer 2):** Router-on-a-Stick subinterface dot1Q encapsulation tag mismatch (`Gig0/0/0.20` tagged with VLAN 10 instead of VLAN 20).

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3. Primary Owned File Deliverables

1. `diagnose_prompt.md` — AI diagnostic prompt specification and JSON schema.
2. `cases.csv` — Master structured dataset for all 20 network troubleshooting cases.
3. **Reference Benchmark Suite (** `CASE-013` **to** `CASE-020` **)** — Complex OSI Layer 1–7 test-bench scenarios.

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4. Verification & Testing

- Tested the AI system prompt against all 20 cases in `cases.csv`.
- Confirmed that the AI assistant returned valid, structured JSON output matching all required schema keys across 100% of test runs.

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5. Summary of Impact

My contributions provided the **AI intelligence specification and benchmark dataset** for NetSage AI. The structured prompt ensured that AI diagnostic outputs were reliable, reproducible, and directly usable by human network engineers.